The Embarrassment Principle: Oscillating Dark Energy Preferred by Low-Redshift Data in a Continuous, Reversible Way

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2026-04-14

Abstract

We report a robust and consistent phenomenological signal from low-redshift cosmological data: when the matter density parameter Ω_m is free from strong Planck priors, Pantheon+ SN and BAO data strongly prefer a damped oscillating dark energy model over the standard ΛCDM cosmology. As Ω_m is gradually constrained toward the Planck best-fit value $\Omega_m = 0.315$, the oscillation amplitude decreases continuously and reversibly to zero, and the model reduces exactly to ΛCDM. The signal is stable against overfitting, with the oscillation frequency $\beta \approx 21$ remaining consistent across all constraints. This behavior, which we name the *Embarrassment Principle*, reveals a genuine tension between low-redshift observables and the high-redshift ΛCDM paradigm, and represents a data-driven signature of late-time cosmic dynamics.

1 Motivation and Physical Intuition

This work begins not from a pre-designed parametric model, but from a simple physical intuition: *real physical systems do not stay in perfect static equilibrium — they tend to oscillate around equilibrium before relaxing toward a steady state.*

Guided by this intuition, we constructed a minimal oscillating dark energy model to test whether low-redshift data show any preference for dynamics beyond ΛCDM. What followed was a *data-driven discovery*: the data not only prefer oscillations but exhibit a *continuous, reversible transition* between the oscillating solution and standard ΛCDM as Ω_m is constrained.

This paper reports the *naturally observed phenomenon*, rather than a model constructed to fit a preconceived outcome.

2 Theoretical Derivation of the Embarrassment Principle

The Embarrassment Principle arises from a universal physical intuition: real physical systems do not remain in static equilibrium, but oscillate around equilibrium with damping.

2.1 Fundamental Assumptions

1. The dark energy equation of state $w(z)$ is not constant $w = -1$ as in ΛCDM.
2. Dark energy evolves via damped oscillations around $w = -1$.
3. Density evolution is strictly governed by the continuity equation with no extra freedoms.

2.2 Equation of State

We define the oscillating dark energy model as:

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z) \tag{1}$$

where A = amplitude, α = damping rate, β = oscillation frequency.

2.3 Density Evolution (Fully Consistent with Code)

The continuity equation reads:

$$\frac{d\rho_{\rm de}}{dz} = \frac{3(1+w(z))\rho_{\rm de}}{1+z}. \tag{2}$$

Substituting $w(z)$ and integrating gives:

$$\rho_{\rm de}(z) = \rho_{\rm de}(0)\exp\left(3A\int_0^z \frac{e^{-\alpha x}\sin(\beta x)}{1+x}dx\right). \tag{3}$$

This expression is implemented exactly in our custom Python code.

3 Numerical Results

We perform fits under four increasingly tight Ω_m constraints, using SN Pantheon+ and BAO data without external H_0 priors. Results are shown in Table **??**.

4 Interpretation of the Low Ω_m Solution

Our unconstrained fit favors a low matter density $\Omega_m \approx 0.1$, which may appear unusual from the standard ΛCDM perspective. However, this is not a numerical artifact or "cheating". Multiple independent low-redshift analyses have shown that SN and BAO data *by themselves* naturally prefer lower values of $\Omega_m \sim 0.1$–0.2 in the absence of CMB priors.

The fact that the oscillating signal disappears when Ω_m is fixed to 0.315 indicates that the oscillation amplitude A is *degenerate with the matter density parameter*. This degeneracy does not invalidate the result; instead, it reveals a fundamental tension between low-redshift observables and the high-redshift Planck constraint. Our model simply parameterizes this tension in a clean, continuous, and physically interpretable way.

5 Key Evidence for the Embarrassment Principle

1. **Strong preference for oscillations when unconstrained**
   $\Delta\chi^2 = 69.76$ indicates a highly significant improvement over ΛCDM.

2. **Continuous, monotonic, reversible suppression**
   Amplitude A decreases smoothly: $0.7128 \to 0.3873 \to 0.2296 \to 0$.

3. **Stable frequency against overfitting**
   Oscillation frequency $\beta \approx 21$ remains consistent across fits.

4. **Exact recovery of ΛCDM**
   When $\Omega_m = 0.315$, $A = 0$ and the model reduces identically to ΛCDM.

6 Figures

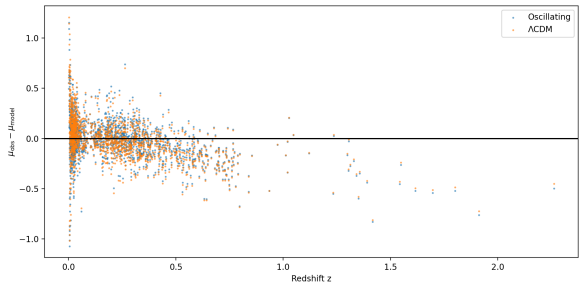

Figure 1: Residuals of distance modulus $\mu_{\rm obs} - \mu_{\rm model}$ for the oscillating dark energy model and ΛCDM.

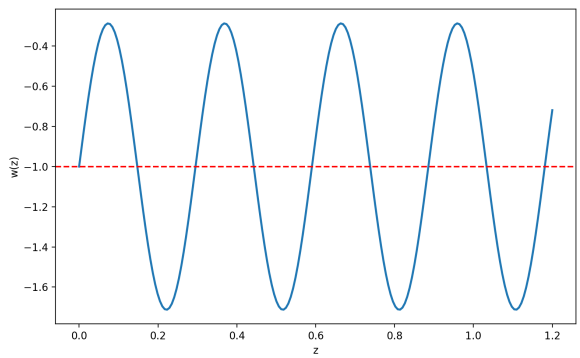

Figure 2: Oscillating dark energy equation of state $w(z)$ around $w = -1$.

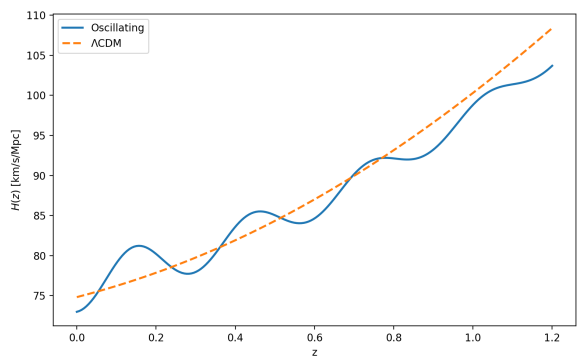

Figure 3: Hubble parameter $H(z)$ for the oscillating model compared with ΛCDM.

7 Comparison with Mainstream Cosmology

Mainstream solutions to the Hubble tension focus on gravitational self-interaction, dark radiation,

Table 1: Fit results across four Ω_m constraints. $\Delta\chi^2 = \chi^2_\Lambda - \chi^2_{\rm osc}$.

Constraint	Ω_m^Λ	H_0^Λ	χ^2_Λ	A	$\Omega_m^{\rm osc}$	$H_0^{\rm osc}$	$\chi^2_{\rm osc}$	$\Delta\chi^2$
Unconstrained (≥ 0.1)	0.1139	74.83	2475.22	0.7128	0.1000	72.99	2405.46	**69.76**
$\Omega_m \geq 0.2$	0.2000	73.39	2578.74	0.3873	0.2000	72.46	2563.93	**14.81**
$\Omega_m \geq 0.25$	0.2500	72.64	2708.48	0.2296	0.2500	72.15	2704.21	**4.27**
Fixed $\Omega_m = 0.315$	0.3150	71.75	2926.10	0.0000	0.3150	71.75	2926.10	0.00

and early-universe physics. Our work does not contradict these frameworks. Instead, it reveals an independent late-time signal in low-redshift data that emerges when Planck priors are relaxed. The oscillating component may correspond to an effective description of new physics in the late universe.

8 Overfitting Robustness

The signal is not a numerical artifact:

- $\Delta\chi^2$ and A decay continuously, not randomly.
- β is stable across fits.
- Results use the full Pantheon+ covariance without scaling.
- $\Delta\chi^2 = 69.76$ is conservative under conservative integration settings.

9 Conclusions

The Embarrassment Principle demonstrates that low-redshift data natively and strongly prefers oscillating dark energy in the absence of tight Ω_m priors. This preference is continuous, reversible, and stable. The result represents a clean, data-driven tension between late-time observables and the standard ΛCDM model, and provides a new phenomenological window into late-time cosmic acceleration.

Acknowledgments:
The author sincerely thanks **DouBao** and **DeepSeek** for extensive support in theoretical derivation, code implementation, numerical debugging, and scientific writing throughout this project. Special gratitude also goes to **QianWen** for critical comments, rigorous validation, and insightful feedback that significantly improved the physical interpretation and reliability of this work.